
PROMPT ENGINEERING REPORT

MediAssist AI

Clinical Decision-Support Prompting with a Hugging Face LLM

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1. Introduction

MediAssist AI is a prompt-engineering project that uses an instruction-tuned Hugging Face language model to build a healthcare assistant. Instead of fine-tuning the model, carefully designed prompts were used to generate safe, structured, and relevant responses. This report explains the model selection, prompt engineering techniques, challenges faced, and improvements made throughout the project.

2. Model Selection

The project uses the Hugging Face HuggingFaceH4/zephyr-7b-beta model to generate responses. The model is loaded using the Transformers library (AutoTokenizer and AutoModelForCausalLM), and a single response function is used to handle all tasks consistently.

Model	Params	Best Use Case	Advantages	Limitations
HuggingFaceH4/zephyr-7b-beta	7B	Instruction-following chat assistant (used for MediAssist AI)	Open-source and freely accessible, follows instructions well, and generates structured responses consistently.	Not medically fine-tuned, can hallucinate; 7B ceiling on complex multi-step reasoning
mistralai/Mistral-7B-Instruct-v0.1	7B	General-purpose instruction tasks; fine-tuning base	Strong base performance; widely benchmarked; efficient	Requires Hugging Face access approval, and its default instruction-following ability is not as strong as Zephyr.

3. Prompt Engineering Techniques Used

- **Role/Persona Prompting:** Different roles such as a clinical assistant, doctor, nurse, or medical data extractor were assigned to the model so it could provide responses suitable for each task.
- **System and User Messages:** Separate system and user prompts were used to clearly define the model's behavior while keeping the user's question independent.
- **Context-Based Prompting:** Additional context was provided in prompts to help the model generate more relevant and accurate responses. For example, the diabetes explanation was written for a newly diagnosed patient.
- **Few-Shot Prompting:** A few input-output examples were included before the actual question so the model could better understand the expected response format.
- **Chain-of-Thought Prompting:** The model was encouraged to think through the problem before answering, while only displaying the final response to the user.

- **Structured Data Extraction:** Clear instructions were used to generate JSON output, and additional parsing logic was included to handle formatting errors if they occurred.
- **Prompt Optimization:** Initial prompts were tested and improved by adding better context, clear instructions, and output constraints, resulting in more accurate and consistent responses.

4. Challenges Faced

- **General-purpose model:** The selected model was not specifically trained for medical tasks, so prompts had to include clear instructions to generate safe and relevant responses.
- **Hallucinated information:** The model could sometimes produce incorrect or made-up medical information. To reduce this, prompts included clear constraints and asked the model to avoid guessing.
- **Generating structured responses:** It was sometimes difficult to get the model to return outputs in the required format, such as Markdown or JSON. More specific prompt instructions helped improve consistency.
- **Balancing detail and clarity:** Some responses were either too long or lacked important information. Adding word limits and a fixed response structure made the outputs more concise and useful.
- **Choosing the right model:** Different Hugging Face models were compared before selecting Zephyr-7B, as it was freely available and followed instructions more effectively for this project. Reasoning, which is why Task 7's optimized prompt explicitly forces a ranked list with a one-sentence rationale per item instead of leaving the reasoning implicit.

5. Improvements Made After Prompt Refinement

5.1 Diabetes explanation (Task 2)

Before: The prompt *"Tell me about diabetes"* was too general and did not specify the audience, response format.

After: The prompt was improved by assigning the role of an endocrinologist, targeting newly diagnosed patients, using simple language, setting a word limit, and organizing the response with clear Markdown headings.

Result: The response became more structured, easier to understand, and more consistent, making it more useful for patients

5.2 Differential diagnosis (Task 7)

Before: The original prompt produced a vague response with multiple possible diseases but no clear reasoning or recommendation.

After: The prompt was improved by providing the patient's symptoms, adding clear instructions, requesting a ranked list of possible conditions, and asking for the recommended next step.

Result: The model generated a more accurate, well-structured, and useful response with clear recommendations instead of uncertain guesses.